

PR Review Summary — v4.0.0 Release

Branch: `release/v4.0.0` → `main`
Commit: `4aed987`
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Date: 2026-05-14

What This PR Does

Implements v4.0.0 of the QPLANT Cryo Leak Rate Dashboard with:

- 1. **Single Source of Truth (SSoT)** — `data/config.yaml` drives all parameters
- 2. **HP Compressor Count Correction** — 4 → **3 units** (per vendor confirmation)
- 3. **Pressure Parameter Updates** — HP discharge: **14 barg**, LP suction: **1.05 bara**
- 4. **Full CI/CD Pipeline** — GitHub Actions for testing, validation, and deployment
- 5. **Recursive Build Tracking** — `BUILD_LOG.md` with SHA-256 provenance
- 6. **Enhanced Navigation** — Hero pages, updated NAVIGATOR, stakeholder presentation
- 7. **Automation Scripts** — Hash computation, link validation, release, deploy, PDF export

Impact Analysis

Metric	Value
Files Changed	39
Lines Added	+2,431
Lines Removed	−210
Net Change	+2,221
Tests	22/22 passing 
HTML Files	84 (all link-checked)
Breaking Changes	Yes — see below

Breaking Changes

Parameter	Old Value	New Value	Reason
HP compressor count	4	3	Vendor (Cryoworld) confirmed 3-unit configuration
HP discharge pressure	16 barg	14 barg	Updated per addendum specifications
LP suction pressure	1.0 bara	1.05 bara	Updated per addendum specifications
Configuration source	Hardcoded	config.yaml	SSoT architecture

Key Changes by Category

Configuration (SSoT)

File	Status	Description
<code>data/config.yaml</code>	New	Single Source of Truth for all parameters
<code>src/config_loader.py</code>	New	YAML → Python config loader
<code>tests/test_config_loader.py</code>	New	Config validation tests

CI/CD Pipeline

File	Status	Description
<code>.github/workflows/ci.yml</code>	New	Test + validate + link-check on PR/push
<code>.github/workflows/deploy.yml</code>	New	Auto-deploy to GitHub Pages on merge
<code>scripts/release.sh</code>	New	Versioned release with SHA-256
<code>scripts/deploy.sh</code>	New	One-command deployment
<code>scripts/compute_hashes.py</code>	New	SHA-256 hash scanner
<code>scripts/validate_links.py</code>	New	HTML broken link checker
<code>scripts/export_pdfs.sh</code>	New	PDF export via weasyprint

Documentation

File	Status	Description
<code>BUILD_LOG.md</code>	New	Recursive build sequence with provenance
<code>QUICK_ACCESS.md</code>	New	URL table for all dashboards
<code>NAVIGATOR.html</code>	Updated	Added hero/CI sections
<code>docs/heroes/executive.html</code>	New	Executive ROI hero page
<code>docs/heroes/technical.html</code>	New	Technical specs hero page
<code>docs/heroes/compliance.html</code>	New	Standards compliance hero page

Deployment

File	Status	Description
<code>netlify.toml</code>	New	Netlify deployment config
<code>vercel.json</code>	New	Vercel deployment config

Testing

File	Status	Description
tests/test_outputs.py	Updated	References updated to v4.0.0
All 7 test files	Pass	22 tests, 100% pass rate

Generated Artifacts

File	Type	Description
BUILD_LOG.docx / .pdf	Export	Build log in Word/PDF
QUICK_ACCESS.docx / .pdf	Export	Quick access in Word/PDF
HANDOVER.docx / .pdf	Export	Handover package
AMBIGUITY_FIXES.docx / .pdf	Export	Ambiguity resolution
Multiple .docx / .pdf pairs	Export	Comprehensive doc suite

Test Results

tests/test_engineering.py		5 passed
tests/test_calculations.py		4 passed
tests/test_config_loader.py		3 passed
tests/test_data_integrity.py		4 passed
tests/test_build_outputs.py		3 passed
tests/test_outputs.py		2 passed
tests/conftest.py		(fixtures)
TOTAL		22/22 passed

Review Checklist for Approvers

Functional

- ☐ CI checks passed (all green)
- ☐ Tests cover new features (config_loader, calculations)
- ☐ Breaking changes documented (compressor count, pressure)
- ☐ SSOT config.yaml is complete and correct

Security & Quality

- ☐ No credentials or secrets in code
- ☐ No hardcoded paths
- ☐ Dependencies pinned in requirements.txt

- [] `.gitignore` covers sensitive files

Documentation

- [] CHANGELOG.md updated
- [] BUILD_LOG.md complete
- [] README current
- [] NAVIGATOR.html navigable

Deployment

- [] CI workflow triggers correctly on PR
- [] Deploy workflow targets correct branch
- [] GitHub Pages source = GitHub Actions



Post-Merge Steps

#	Action	Command/URL
1	Monitor Actions	<code>https://github.com/GB0GEB/cryo_leak_rate_dashboard/actions</code>
2	Verify Pages	<code>https://gbogeb.github.io/cryo_leak_rate_dashboard/</code>
3	Test live URLs	See <code>QUICK_ACCESS.md</code>
4	Create git tag	<code>git tag -a v4.0.0 -m "Release v4.0.0"</code>
5	Push tag	<code>git push origin v4.0.0</code>
6	Create Release	GitHub → Releases → New Release
7	Run validation	<code>./scripts/post_merge_validation.sh</code>



Related Documents

- [PR_CHECKLIST.md](#) (PR_CHECKLIST.md) — Detailed merge checklist
- [GITHUB_INTEGRATION_STATUS.md](#) (GITHUB_INTEGRATION_STATUS.md) — Full status report
- [GITHUB_PAGES_SETUP.md](#) (GITHUB_PAGES_SETUP.md) — Pages configuration guide
- [BUILD_LOG.md](#) (BUILD_LOG.md) — Complete build provenance
- [QUICK_ACCESS.md](#) (QUICK_ACCESS.md) — All dashboard URLs

